

Credit Risk Intelligence

Project Documentation

Group DomainRange

UIU Data Analytics Laboratory

Musfique Ahmed · Tasfiya Binte Karim

```
application_train.csv
```

Approve /

Intelligence

```
src/data/clean.py reports/data_quality_report.md reports/cleaning_summary.md
```

Intelligence

DAYS_EMPLOYED == 365243 NaN

COMMONAREA *

LIVINGAPARTMENTS * YEARS BUILD *

' MISSING '

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OCCUPATION_TYPE	ORGANIZATION_TYPE
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StandardScaler

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Musique_Ahmed_Tasfiya_Rinto_Karim
data/processed/train_clean.parquet

```
notebooks/01_eda.ipynb
```

```
reports/figures/01..12_*.png
```

3.2 Three hypothesis tests

hypothesis tests

ID	Hypothesis	Result
H1	Lower EXT_SOURCE → higher default	■ — $r \approx -0.16$ to -0.18 , $p \approx 0$ SUPPORT
H2	Higher credit-to-income ratio → higher default	■ It— inverted-U shape; Simpson's paradox
H3	Occupation/education segments have elevated risk	■ Low-skill Laborers 17.2% vs Accountants 1.2%

3.3 Key findings (with numbers)

EXT_SOURCE_3 is the single most important external source of information for creditworthiness (CT) and is followed by EMPLOYED_YEARS and CREDIT_GOODS_RATIO. The importance of these features is not surprising, as they are all related to the borrower's financial situation and ability to repay. The importance of EMPLOYED_YEARS is particularly high, as it is a strong indicator of the borrower's stability and income. The importance of CREDIT_GOODS_RATIO is also high, as it is a key factor in determining the borrower's creditworthiness. The importance of EXT_SOURCE_3 is also high, as it is a key factor in determining the borrower's creditworthiness. The importance of these features is not surprising, as they are all related to the borrower's financial situation and ability to repay. The importance of EMPLOYED_YEARS is particularly high, as it is a strong indicator of the borrower's stability and income. The importance of CREDIT_GOODS_RATIO is also high, as it is a key factor in determining the borrower's creditworthiness. The importance of EXT_SOURCE_3 is also high, as it is a key factor in determining the borrower's creditworthiness.

3.4 Caveats

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src/features/engineer.py src/features/select.py notebooks/02_feature_selection.ipynb reports/feature_importance.md

4.1 Engineering Features

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AGE_YEARS	DAYS_BIRTH / -365
EMPLOYED_YEARS	DAYS_EMPLOYED / -365
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CREDIT_INCOME_RATIO	AMT_CREDIT / AMT_INCOME_TOTAL
ANNUITY_INCOME_RATIO	AMT_ANNUITY / AMT_INCOME_TOTAL
CREDIT_GOODS_RATIO	AMT_CREDIT / AMT_GOODS_PRICE

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4.2 Top-30 feature selection

Ranking features by importance and mean rank combined

Rank	Feature	Mean rank
1	EXT_SOURCE_MEAN	1.0
2	EXT_SOURCE_1	1.0
3	EMPLOYED_YEARS	1.0
4	EXT_SOURCE_3	1.0
5	CODE_GENDER_M	1.0

4.3 Baseline vs. reduced comparison

Home Credit Default Risk · 307,511 loans · XGBoost AUC 0.7578

Baseline model: Logistic Regression
Reduced model: XGBoost

Home Credit Default Risk

src/models/train.py notebooks/03_model_building.ipynb
reports/phase4_model_comparison.md

5.1 Why these 4 models?

Model	Family	Why we chose it
Logistic Regression	Linear baseline	Interpretable, fast, gives a coefficient baseline. Necessary for
Random Forest	Bagged trees	Strong out-of-the-box baseline; handles NaN natively; gives fe
XGBoost	Gradient boosting	Industry standard for tabular data; known to win Kaggle-style
MLP (PyTorch)	Deep learning	Provides the DL family. PyTorch chosen because TensorFlow

5.2 Imbalance strategy decision

Family	class_weight='balanced'	SAFARI
Logistic Regression	0.7398	0.7394
XGBoost	0.7539	0.7203

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5.3 Final 4 model comparison (validation)

Model	Imbalance	Config	Val AUC	Precision	Recall	F1
XGBoost	balanced	n_est=3000, lr=0.075, depth=4, scale_pos_weight=11.39	0.7539	0.4154	0.675	0.284
Random Forest	balanced	n_est=3000, depth=None	0.7469	0.4287	0.394	0.286
MLP (PyTorch)	pos_weight	hidden=[128,64], dropout=0.3, epochs=12	0.7448	0.4153	0.3684	0.250
Logistic Regression	balanced	C=10.0	0.7388	0.4156	0.660	0.252

5.4 Outcomes

models/best_model.pkl

5.5 Recommendation policy

src/models/score.py

Probability of default	Recommendation
$p < 0.2$	(clearly low risk) Approve
$0.2 \leq p < 0.5$	(gray zone — human decides) Manual Review
$p \geq 0.5$	(clearly high risk) Reject

Home Credit Default Risk · 307,511 loans · XGBoost AUC 0.7578

5.6 Three synthetic applicant demos

Profile	Probability	Recommendation
Low-risk	0.0073	Approve
Mid-risk	0.3865	Manual Review
High-risk	0.9548	Reject

Source code: `dashboard/app.py` `dashboard/README.md`

Access Streamlit app locally: `http://localhost:8501`

6.1 Tabs

Tab	What it does
Portfolio Overview	KPI cards (215,257 apps, 8.07% default rate, 17,377 defaults) + class balance + 4
Applicant Risk Scorer	14-widget form → live XGBoost score → hero card (Approve / Manual Review / Rej
Feature Importance Explorer	Top-30 features with category filter (EXT_SOURCE / Time / One-hot / Ratio / Amou
Segment Analysis	Default rate by income decile / employment bucket / region / occupation with 95% C

6.2 Design

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Color palette: `#0A1428` `#00D9B5` `#3B82F6` `#F87171`

Theme file: `dashboard/_theme.py`

6.3 Stack note

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Deployment instructions: `>=1.38, <1.50` `conda create --name data-science python=3.8`

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7.1 Tests (23 passing)

File	Tests	What it covers
tests/test_clean.py	3	Phase 1 cleaning + scaler-fit-on-train-only leakage check
tests/test_engineer.py	6	Phase 3 feature engineering edge cases (partial-NaN safety)
tests/test_score.py	8	Phase 4 scoring function schema and policy thresholds
tests/test_dashboard.py	6	Phase 5 in-process Streamlit AppTest smoke test

7.2 Reproduction pipeline

8.1 Quantitative outcomes

Model parameters: `EXT_SOURCE_MEAN` `EXT_SOURCE_1` `EMPLOYED_YEARS` `EXT_SOURCE_3` `CODE_GENDER_M`

Model configuration: `class_weight='balanced'`

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8.2 Qualitative outcomes

Score application function: `score_application(input: dict) -> dict`

Documentation files: `README.md` `dashboard/README.md` `models/README.md`

Project documentation: `docs/PHASE_CHECKLIST.md` `docs/PROJECT_DOCUMENTATION.md`

8.3 Business takeaways

- Lending and credit applications should not be automated. `EXT_SOURCE_3` should be a required field.
- Credit risk applications are also high-risk and high-cost. Credit risk models need to capture the noise. Avoid simple cut-off thresholds.
- Lending should be done with individual features rather than serve as the sole decision rule. Segment-level underwriting is defensible.
- In credit risk applications, income verification should be prioritized.

Credit Risk Intelligence

- Credit Risk Intelligence Sources
- Credit Risk Group
- Credit Risk Data Sources and Data Members
- Credit Risk Data Sources and Data Members
- Credit Risk Data Sources and Data Members

Project Doc `puku_master_prompt.md`

The following are the files and folders in the `docs/PHASE_CHECKLIST.md`

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